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Design and Development of a Robotic Solution for Water Surface Pollution Control

Bikas Mondal, Sanghamitra Layek, Anirban Saha, Devdippro Bhaduri, Somshubhra Bose, Anuska Bharati and Sayan Bag

Abstract

Pollution of water is a serious global problem, impacting the ecosystem and human life. Traditional cleaning is inefficient and time-consuming. This paper introduces a robotic water surface pollution control system with the use of automation, waste sorting, and real-time monitoring. The bot features IoT-based water quality monitoring with pH, TDS, and turbidity sensors and an inductive proximity sensor for effective waste sorting. The bot powers LoRa connectivity and solar power, thus improving sustainability and cost-effectiveness. The system features an automated conveyor belt to effectively gather and sort waste and offer real-time water quality data for timely action. Experimental results confirm its effectiveness, scalability, and potential for long-term environmental conservation. The use of automation, IoT, and renewable energy makes this a viable solution for keeping aquatic life cleaner and healthier.

Keywords: Water pollution, autonomous robots, waste segregation, water quality monitoring, lora communication

Introduction

Urbanization and industry cause water pollution, leading to the accumulation of harmful pollutants in water bodies. Conventional cleaning is not sufficient, and new robotic systems are needed. Autonomous water-cleaning robots are self-sustaining and integrate trash collection, sorting, and real-time water quality testing, with minimal manpower and maximum sustainability. This paper presents a new robotic system for surface water pollution prevention with automated trash picking, IoT-monitoring, and renewable energy integration. With pollutant sensors and inductive proximity-based sortation, the system is effective and scalable for pollution control.

Water pollution is a critical environmental issue, and conventional cleaning technologies have proven ineffective for large-scale applications ^[1]. Robotics advances have created autonomous and semi-autonomous water-cleaning robots featuring automated trash collection and real-time sensing ^[2, 3]. AI-powered trash sorting has improved pollutant classification accuracy and reduced operating costs ^[4], while LoRa-based connectivity enhances robot navigation and data acquisition ^[5]. IoT-based monitoring devices enable real-time monitoring of water quality parameters like turbidity, pH,

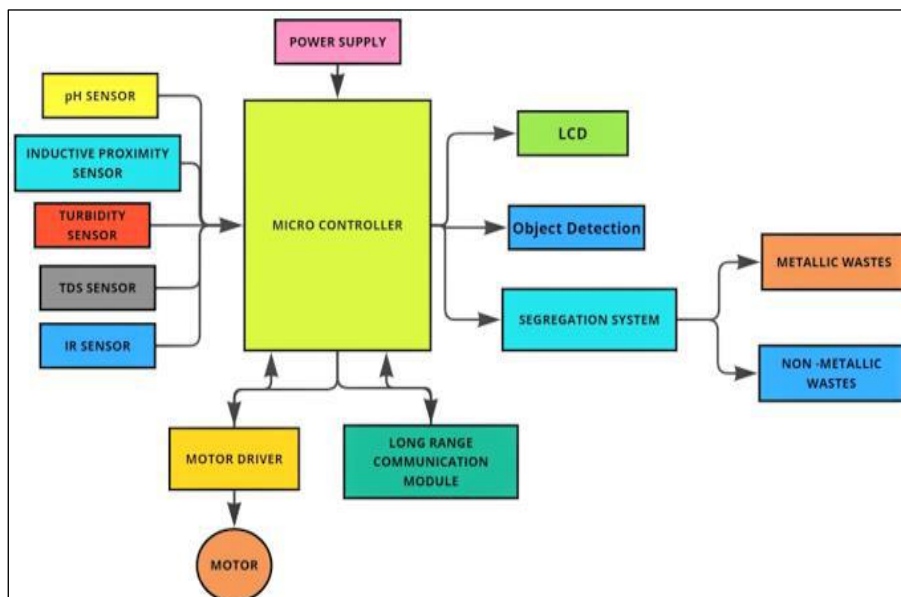


Fig 1: Block diagram of the proposed Bot